MILITARY SPECIFICATION

TELESCOPE, STRAIGHT: XM128

1. SCOPE

1.1 This specification covers a straight telescope of one type, designated the XM128.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitations for bids or requests for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Frankford Arsenal

F8205096	Telescope, Straight: XM128
IEL 10540459	List of Inspection Documents for XM128 Telescope Straight

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PACKAGING DATA SHEETS

10540459 Packaging of Telescope, Straight: XM128

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The telescope shall be manufactured in accordance with Drawing F8205096 and drawings pertaining thereto.

3.1.1 Initial production sample.- Requirements for the submission of an initial production sample by the contractor shall be as specified in the contract.

3.2 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surfaces finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental service conditions.

3.3.1 Storage temperatures.- The telescope shall show no evidence of physical failure such as glass breakage, bonding failure, or casting failure due to storage at temperatures of -80° and +160°F.

3.3.2 Standard ambient temperatures.- The telescope shall meet the requirements of 3.3.3 through 3.7 at a standard ambient temperature of 75 ± 15°F.

3.3.3 Vibration.- The telescope shall be vibrated at a constant frequency of 30 ± 5 cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion). Subsequent to vibration, there shall be no evidence of damage or physical failure and the telescope shall meet the requirements of 3.4 to 3.7 inclusive.

3.4 Cleanliness.- Subsequent to vibration, the telescope shall meet the cleanliness requirements of MIL-0-13830. There shall be no more than 2 particles of dirt visible on the reticle. The area of each particle shall not be in excess of the area equivalent to a size 5 dig. There shall be no other dirt within the telescope which interferes with observation or adversely affects image quality.

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3.5 Sealing.- The telescope shall show no evidence of leakage when subjected to an external pressure of 5 ± 0.10 pounds per square inch gage pressure (psig).

3.6 Optical.

3.6.1 Collimation.- The optical axis of the telescope (through the geometric axis of the reticle) shall be parallel to the bottom of the dovetail mounting surface within 0.5 mil.

3.6.1.1 Open sight.- The line of sight of the open sight shall be parallel with the optical axis of the telescope within one degree.

3.6.2 Image tilt.- With the flat of the dovetail horizontal, the image of a vertical target line shall be vertical within 1 degree of arc.

3.6.3 Parallelism of reticle and image.- With the flat of the dovetail horizontal, the vertical reticle line through the geometric axis shall be parallel to the image of a vertical target line within 30 minutes of arc.

3.6.4 Parallax.- Parallax at the optical axis shall not exceed 0.1 mil when viewing a target at a distance of between 200 meters and infinity.

3.6.5 Resolution.- Resolution on the optical axis shall be 10 seconds of arc or less. Resolution consists of detectable line structure and proper line count in all four meridians within .25 diopters measured at the eyepiece using an auxiliary telescope.

3.6.6 Eyepiece focus.- The eyepiece focus shall be fixed between -0.75 and -1.0 diopter.

3.6.7 Illumination.- The telescope shall meet the "Illuminated reticles" requirement of MIL-o-13830. With the reticle illuminated, the reticle marking shall appear clearly defined when observed in a darkened area.

3.7 Eyeshield.- The rubber eyeshield shall show no evidence of deterioration, cuts, or tears and shall fit snugly on the telescope body.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes, and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.1 Initial production sample failure.- Should one unit fail to meet the initial production testing specified in 4.2, the contractor shall cease those operations which are causing the rejection and make the necessary corrections thereto. When corrections in the process or controls have been made, two additional (new) production models shall be submitted in accordance with 4.2.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examinations and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL 10540459. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and III herein. Examinations and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 1.0% Defective		
101.	Vibration	3.3.3	4.6.1.2
102.	Cleanliness	3.4	4.6.2

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103.	Collimation	3.6.1	4.6.4
104.	Open sight	3.6.1.1	4.6.4.1
105.	Image tilt	3.6.2	4.6.5
106.	Parallelism of reticle and image	3.6.3	4.6.6
107.	Parallax	3.6.4	4.6.7
108.	Resolution	3.6.5	4.6.8
109.	Eyepiece focus	3.6.6	4.6.9
110.	Illumination	3.6.7	4.6.10
111.	Eyeshield	3.7	4.6.11

MINOR: NONE

NOTE: The tests of Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One telescope shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests listed Table II.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental.- Three telescopes shall be selected at rondom as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests listed in Table I and shall then meet the requirements and tests listed in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° and +160°F)	3.3.1	4.6.1.1
304.	Sealing (+60 to 90°F)	3.5	4.6.3

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product

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shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number inspecifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in IEL 10540459.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test methods and procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- Special pressure testing equipment, as shown on Drawing F7680673, shall be capable of being rendered pressure-tight and of holding the required pressure for the specific period of time with a test specimen inserted in the chamber.

4.5.2.3.2 Performance.- Special test equipment utilized to test the telescope shall conform to Drawing F7694010 (See IEL 10540459). The test equipment shall permit inserting and securing the dovetail of the telescope in a horizontal plane within 10 seconds (0°0'10"). The fixture shall also consist of a target collimator containing a calibrated reticle to check collimation, parallax, and accuracy readings.

4.6 Test methods and procedures.

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4.6.1 Environmental.

4.6.1.1 Storage temperatures.- Test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The telescope shall be exposed to each specified temperature extreme for a minimum of five hours. Subsequent to the temperature exposures, the telescope shall be stabilized in an ambient temperature between +60° and 90°F and subjected to the visual, tactile, and audible examination of 4.6.4 to 4.6.11. The telescope shall conform to the requirements of 3.3.1. Caution shall be exercised during this test to avoid subjecting the telescope to thermal shock.

4.6.1.2 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.3.3 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The telescope shall be mounted horizontally by means of an adapter that is secured to the vibration fixture. This adapter shall contain the necessary dovetail slot capable of mounting and securing the dovetail of the telescope during the vibration test. The telescope shall be vibrated in the vertical plane for a period of 5 minutes ± 15 seconds. Subsequent to vibration, the telescope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.3.3 prior to being subjected to the tests specified herein. A visual examination of the telescope shall be made for missing or damaged parts and for chipped or loose paint. Minimum visual examination shall include the name plate, eyeshield and ring, sunshade, dovetail mounting surfaces, screws, washers, pins, sealant, objective lens, eyelens, and reticle illumination window. Optical characteristics shall be examined by viewing through the objective lens and the eyelens of the telescope to detect loose dirt, scratches, digs, fractures, cement separations or foreign matter which would interfere with the optical system and be cause for rejection. A tactile examination shall be conducted to determine looseness of external parts, such as cover, shoe, retaining rings, screws, etc. An audible examination shall also be made for loose internal parts or broken optical components, such as lenses, prisms, reticle, retaining rings, etc. Any one of the possible defects defined above shall be cause to consider a telescope defective, unless separate defects-per-hundred type sampling plans are established and approved by the Government as part of the supplier's inspection system. Telescopes which have passed the above examinations shall be subjected to the tests of 4.6.2 to 4.7 inclusive to determine overall compliance with 3.3.2.

4.6.2 Cleanliness.- The optical surfaces of the telescope shall be inspected for cleanliness in accordance with MIL-O-13830. The telescope shall conform to the cleanliness requirements of 3.4.

4.6.3 Sealing.- The external pressure testing of the telescope shall be performed with special test equipment conforming to 4.5.2.3.1. The pressure shall be applied to the telescope by means of the adapter specified in the test equipment. Place the telescope in the adapter and pressurize with dry compressed air. Retain the applied pressure of 5 ± 0.10 pounds per square inch gage (psig) on the telescope for a period of 5 minutes ±

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15 seconds. At the end of the time period there shall be no evidence of leakage as detected on the dial indicators of the testing equipment, to conform to the requirement of 3.5.

4.6.4 Collimation.- The collimation test is accomplished by mounting the telescope on the fixture (see 4.5.2.3.2), against the stop, and securing it with the thumbnut and gib. Sight through the telescope eyepiece and observe the coincidence between the target center point and the intersection at the top of the vertical dash line at the center of the horizontal 1400 yard range line. Both points must coincide within the tolerance specified in 3.6.1.

4.6.4.1 Open sight.- The accuracy of the open sight shall be checked on a wall target representing the limit specified in 3.6.1.1. The size of the checking target will be determined by the distance the target is located from the objective of the telescope under test. The telescope shall be removed from the holding device on the fixture and secured in a similar device to make this test.

4.6.5 Image tilt.- This test shall be performed with the aid of an auxiliary telescope containing a vertical and horizontal reticle line, and the test equipment in 4.5.2.3.2. The telescope shall be at least 3 power (3X). Plumb the vertical reticle line of the telescope to a plumb line and position at the eyepiece of the telescope under test. View through the telescope and establish one extremity of the telescope vertical reticle line coincident to the respective target image line. The image of the target line shall be parallel to the telescope's reticle line within the tolerance specified in 3.6.2 as measured on the target.

4.6.6 Parallelism of reticle and image.- This test is made with one extremity of the vertical reticle line of the auxiliary telescope in coincidence with the vertical target line image. The reticle of the telescope under test shall be parallel to the vertical target line within the tolerances specified in 3.6.3 as measured on the target.

4.6.7 Parallax.- This test shall be performed by observing through the telescope eyepiece. Parallax is recognized as displacement caused between the telescope reticle and the image of the target when the observer's head is moved from side to side or up and down. Such movement of the observer's head shall not cause displacement of the two images to exceed the tolerance specified in 3.6.4 at the distance indicated.

4.6.8 Resolution.- A resolution target and an auxiliary telescope of at least 3 power (3X) shall be used for this test. The resolution target shall be a resolving power chart on the wall target, representing the angular subtense for seconds of arc specified in 3.6.5. Fabrication of the resolving power chart shall be as specified in the "Resolution test" of MIL-O-13830, and shall contain four sets of lines as shown in Figure 1 therein. The target shall be illuminated with white light. Position the telescope at the eyepiece of the telescope under test and view the resolving power chart. It shall have the detectable line structure and proper line count in each meridian. The telescope shall resolve the target within the requirements of 3.6.5.

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4.6.9 Eyepiece focus.- This test shall be performed utilizing a dioptometer with at least a minus 1.0 diopter to a plus 1.0 diopter setting. The crossline reticle of the dioptometer shall be adjusted to the individual eye. Placing the dioptometer at the eyepiece of the telescope under test, observe the sharpness of the telescope reticle in comparison to the reticle in the dioptometer. Adjust the dioptometer until both reticle lines appear to be equal, then read the diopter scale of the dioptometer. The index shall be within the tolerance specified in 3.6.6.

4.6.10 Illumination.- The illumination of the graduations on the reticle shall be accomplished with a light source adapted to the illumination part of the telescope and energized by a 1-1/2 to 2 volt energy supply. This is a visual check performed in a darkened area. The graduations shall appear to be evenly illuminated without flaring at the edge of the graduations or bright spots over the reticle area as required by 3.6.7.

4.6.11 Eyeshield.- The eyeshield shall be given a visual and tactile examination. The material shall show no physical damage or deterioration and it shall fit snug to the eyepiece end of the telescope as required by 3.7.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10540459. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This telescope is intended for use in conjunction with Radar Chronograph: M36 for alinement purposes.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

Custodian:

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Preparing activity:

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Project No. 1290-A209

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